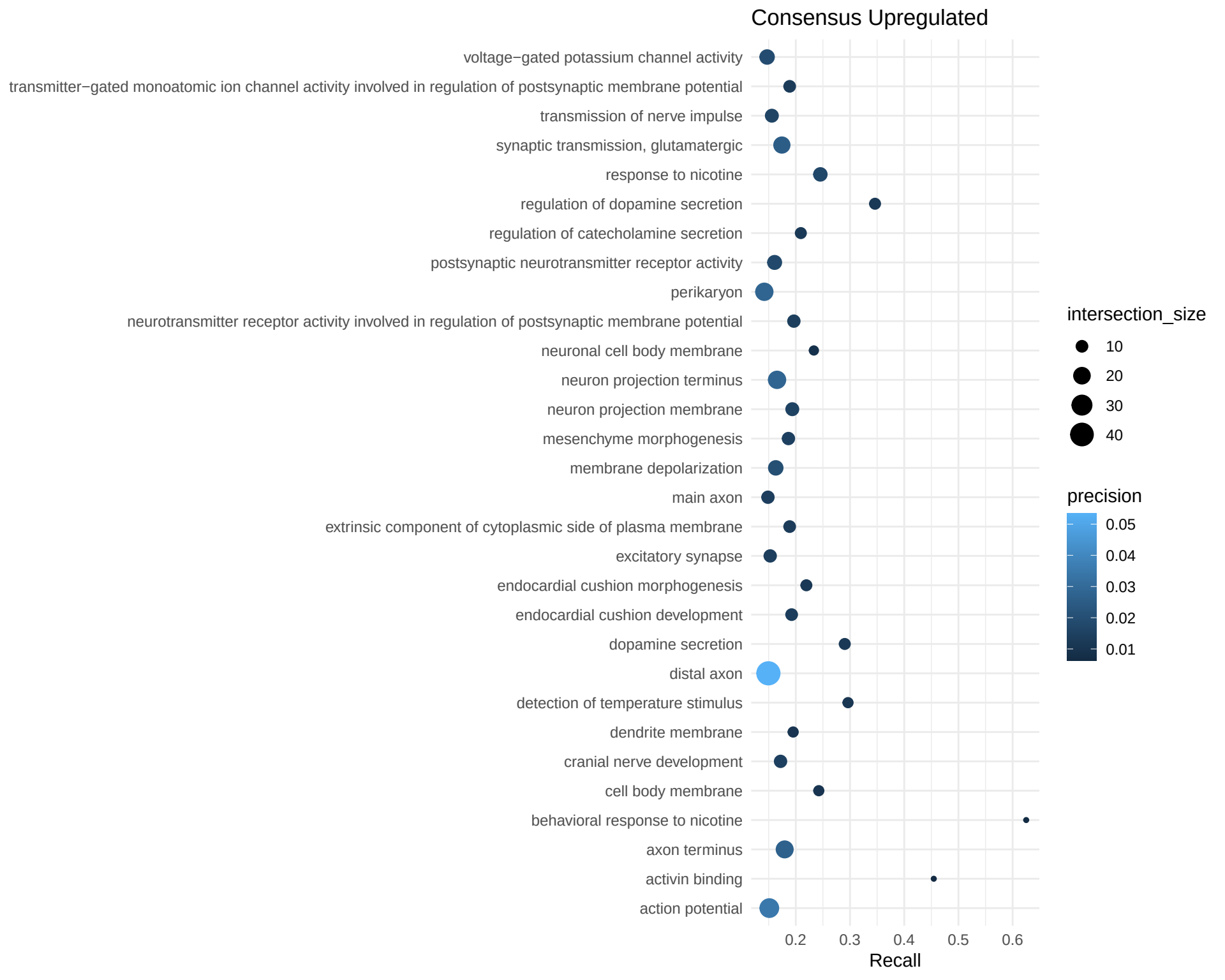
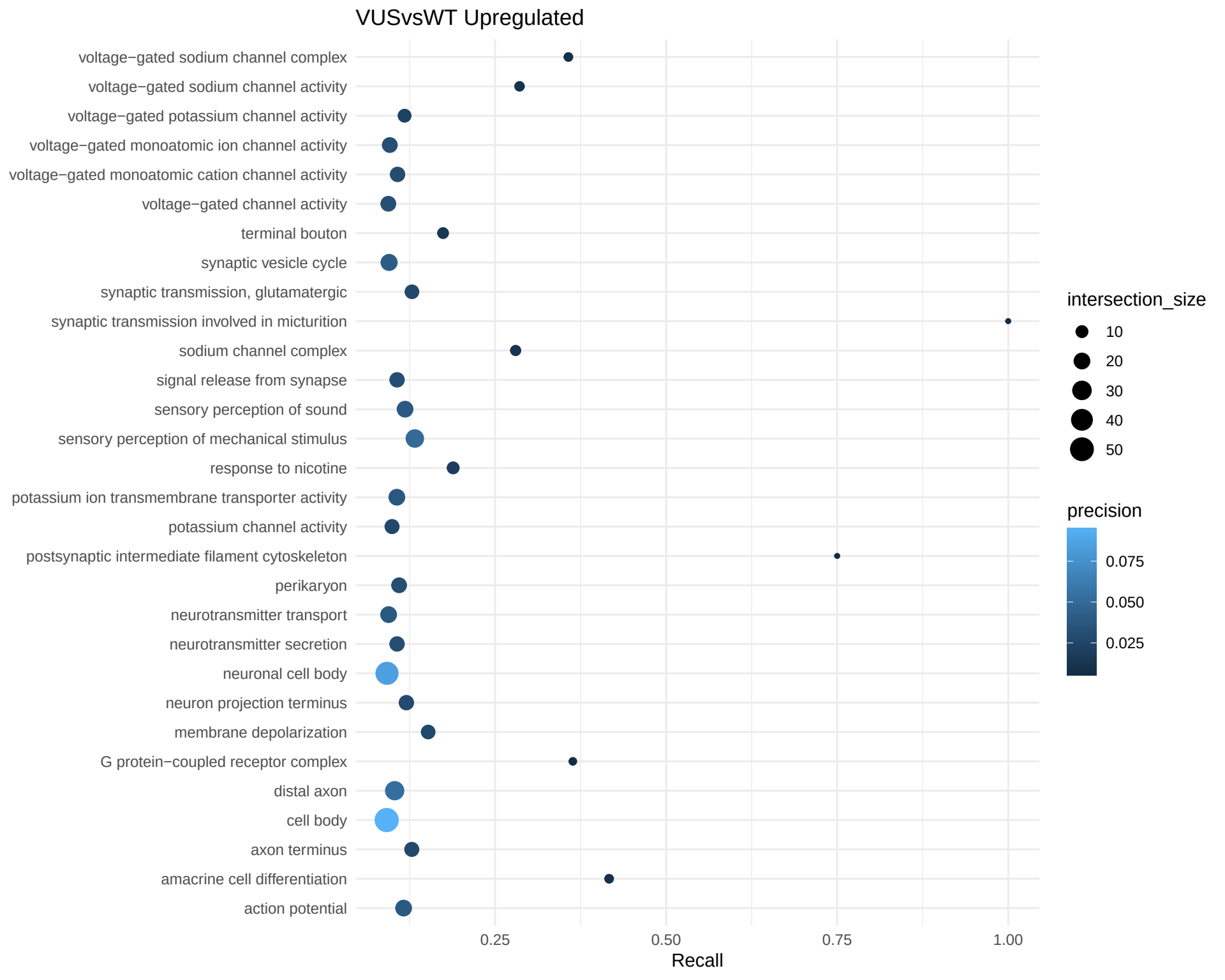
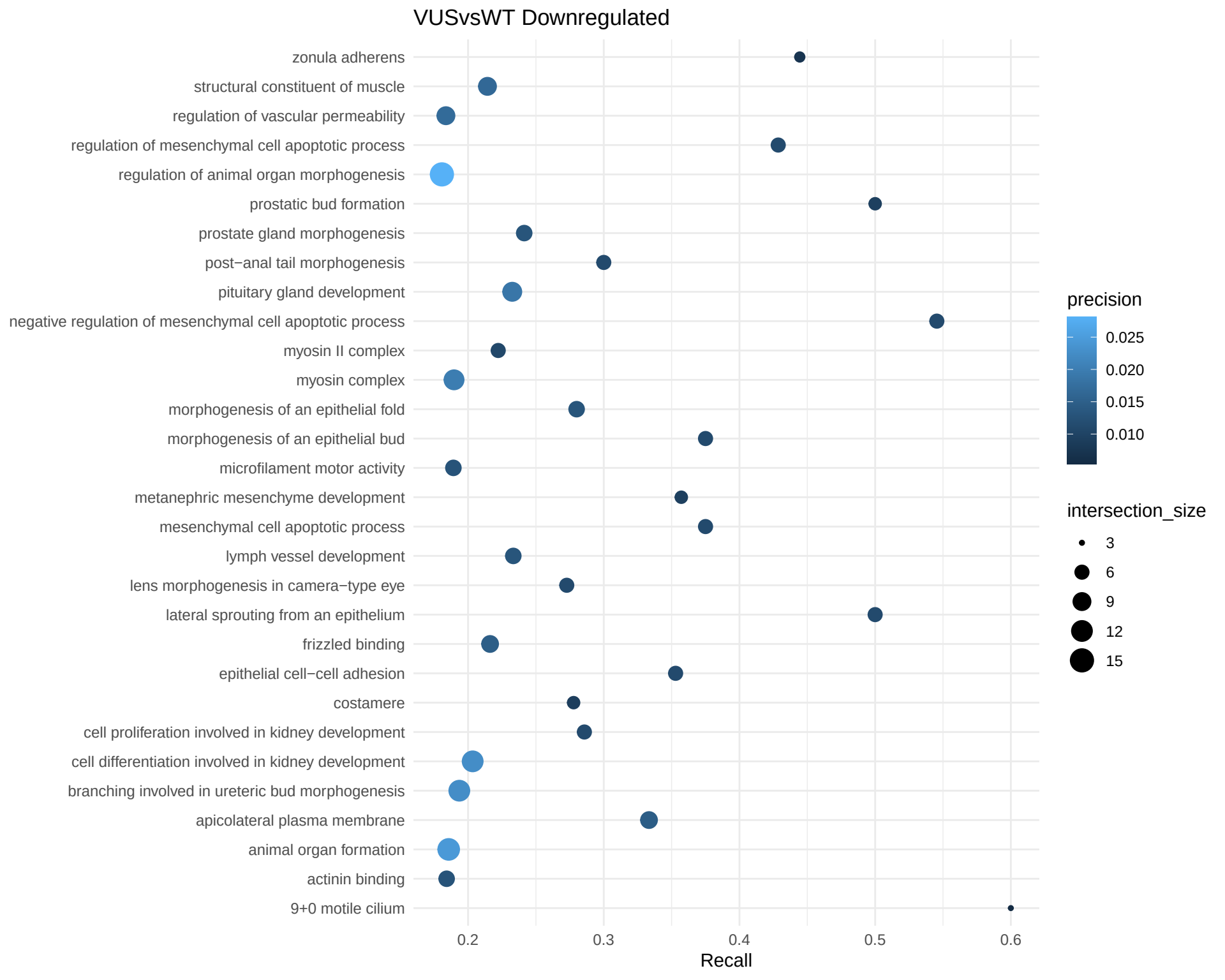
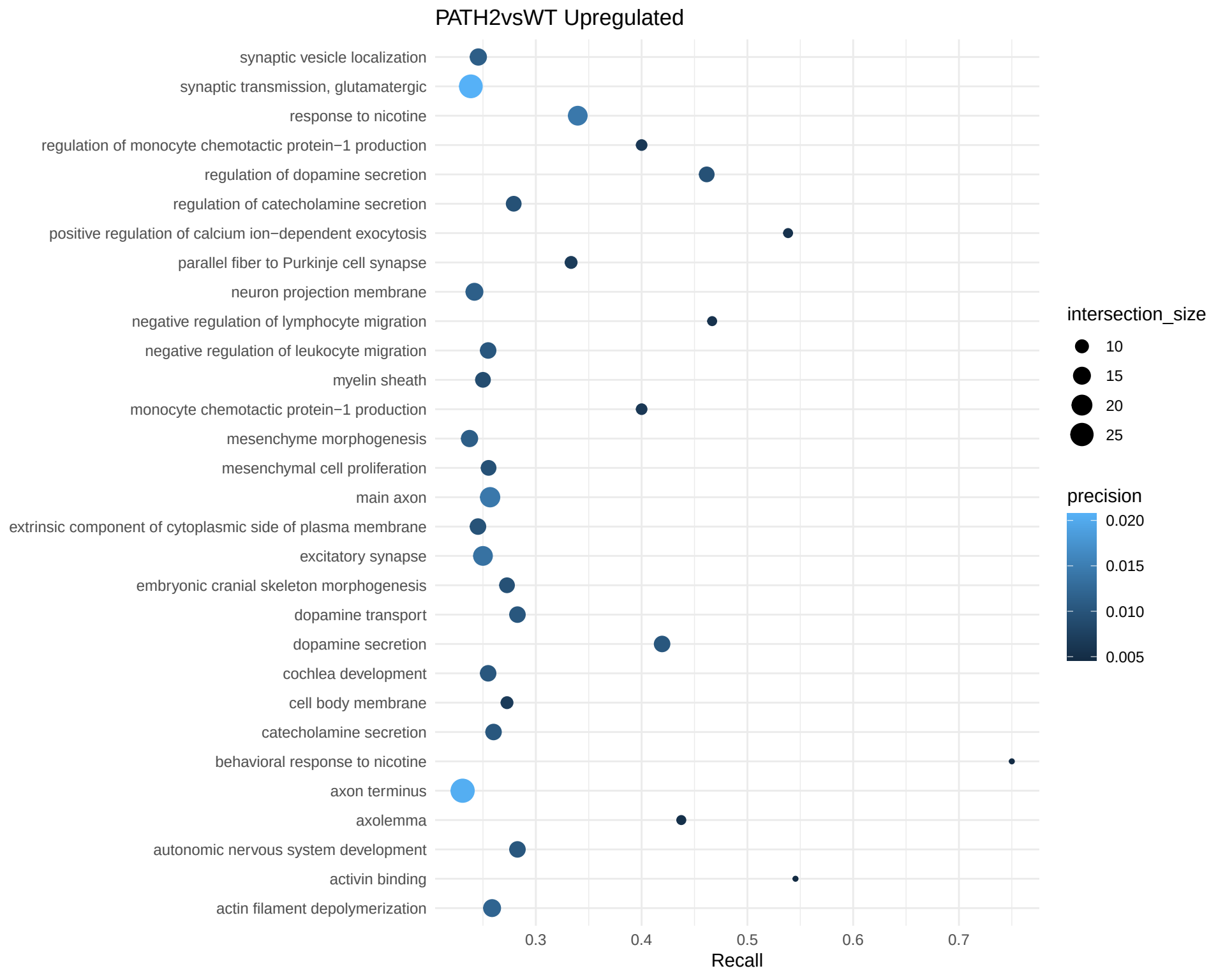
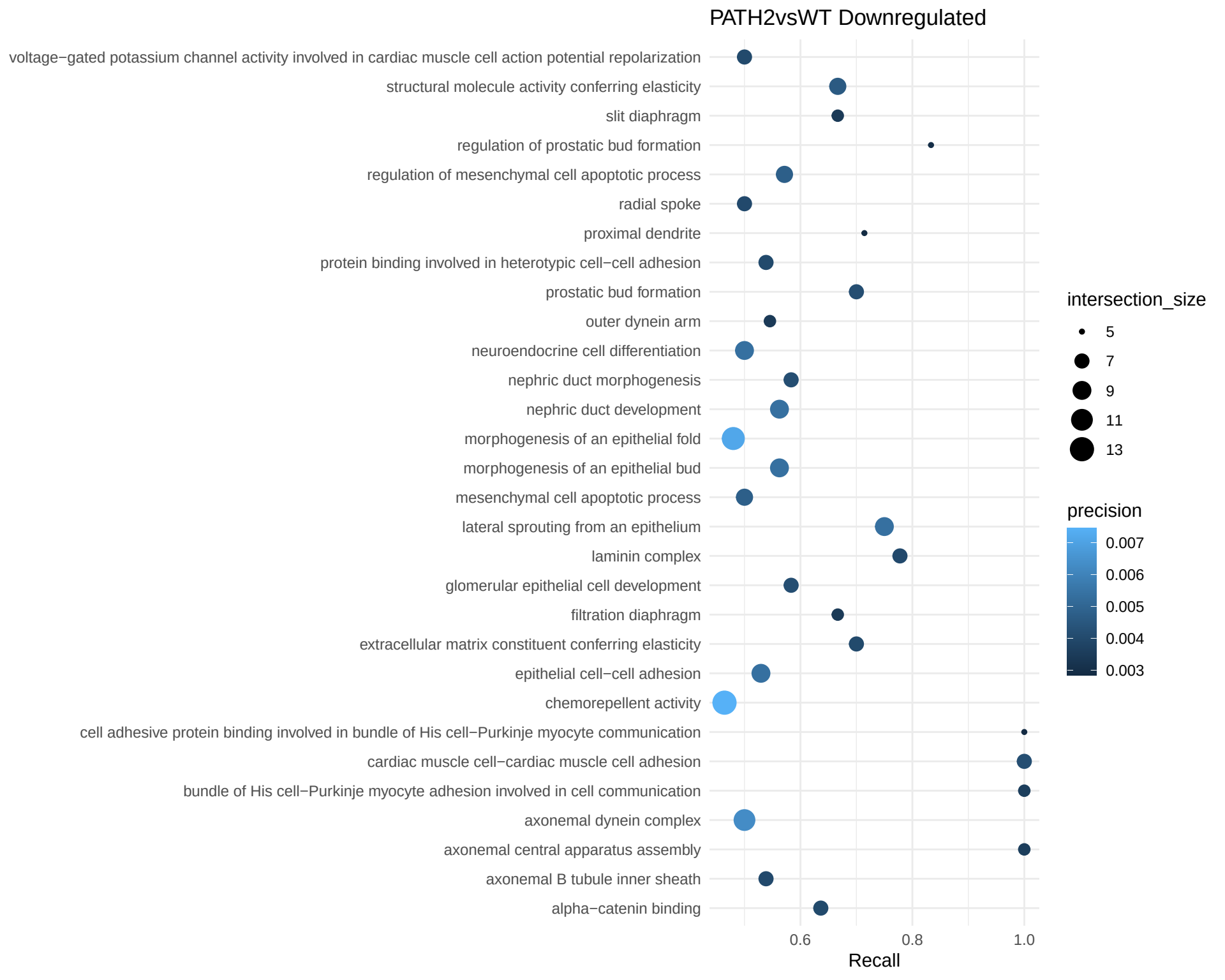

Consensus Upregulated
voltage−gated potassium channel activity
transmitter−gated monoatomic ion channel activity involved in regulation of postsynaptic membrane potential
transmission of nerve impulse
synaptic transmission, glutamatergic
response to nicotine
regulation of dopamine secretion
regulation of catecholamine secretion
postsynaptic neurotransmitter receptor activity
perikaryon
neurotransmitter receptor activity involved in regulation of postsynaptic membrane potential
neuronal cell body membrane
neuron projection terminus
neuron projection membrane
mesenchyme morphogenesis
membrane depolarization
main axon
extrinsic component of cytoplasmic side of plasma membrane
excitatory synapse
endocardial cushion morphogenesis
endocardial cushion development
dopamine secretion
distal axon
detection of temperature stimulus
dendrite membrane
cranial nerve development
cell body membrane
behavioral response to nicotine
axon terminus
activin binding
action potential
0.2
0.3
0.4
0.5
0.6
Recall
intersection_size
10
20
30
40
precision
0.05
0.04
0.03
0.02
0.01

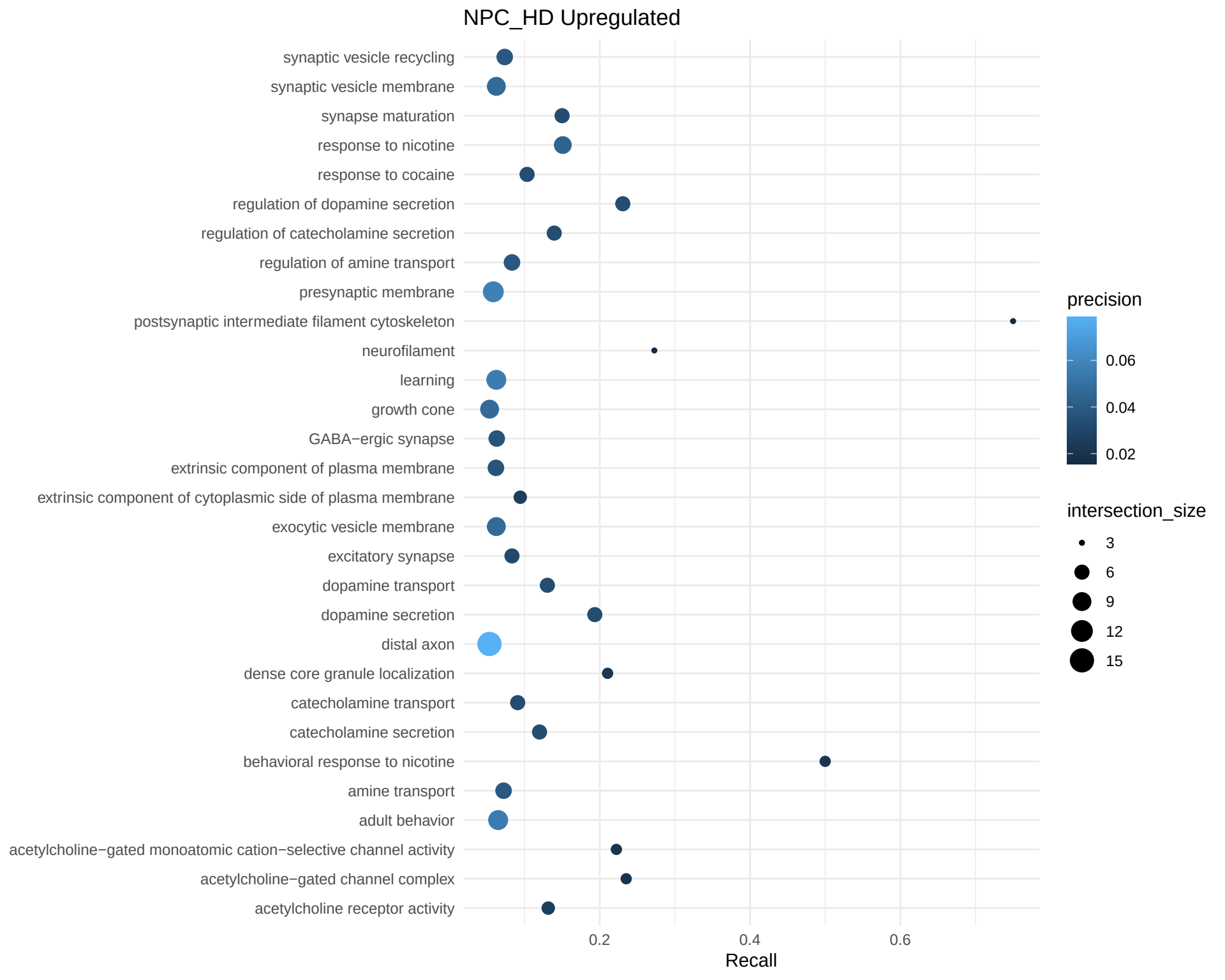
Consensus Downregulated
striated muscle thin filament
specification of animal organ identity
salivary gland morphogenesis
regulation of odontogenesis
regulation of mesenchymal cell apoptotic process
regulation of cardiocyte differentiation
regulation of animal organ formation
protein localization to cell−cell junction
outflow tract septum morphogenesis
nucleoside monophosphate kinase activity
neuroendocrine cell differentiation
nephric duct morphogenesis
nephric duct development
negative regulation of mesenchymal cell apoptotic process
nail development
morphogenesis of an epithelial fold
morphogenesis of an epithelial bud
mesenchymal cell apoptotic process
maintenance of blood−brain barrier
lens morphogenesis in camera−type eye
lateral sprouting from an epithelium
laminin complex
glomerular epithelial cell differentiation
embryonic nail plate morphogenesis
cardiac Troponin complex
cardiac neural crest cell differentiation involved in heart development
cardiac neural crest cell development involved in outflow tract morphogenesis
cardiac neural crest cell development involved in heart development
cardiac muscle cell−cardiac muscle cell adhesion
apicolateral plasma membrane
0.4
0.6
0.8
1.0
Recall
intersection_size
5.0
7.5
10.0
12.5
precision
0.0100
0.0075
0.0050

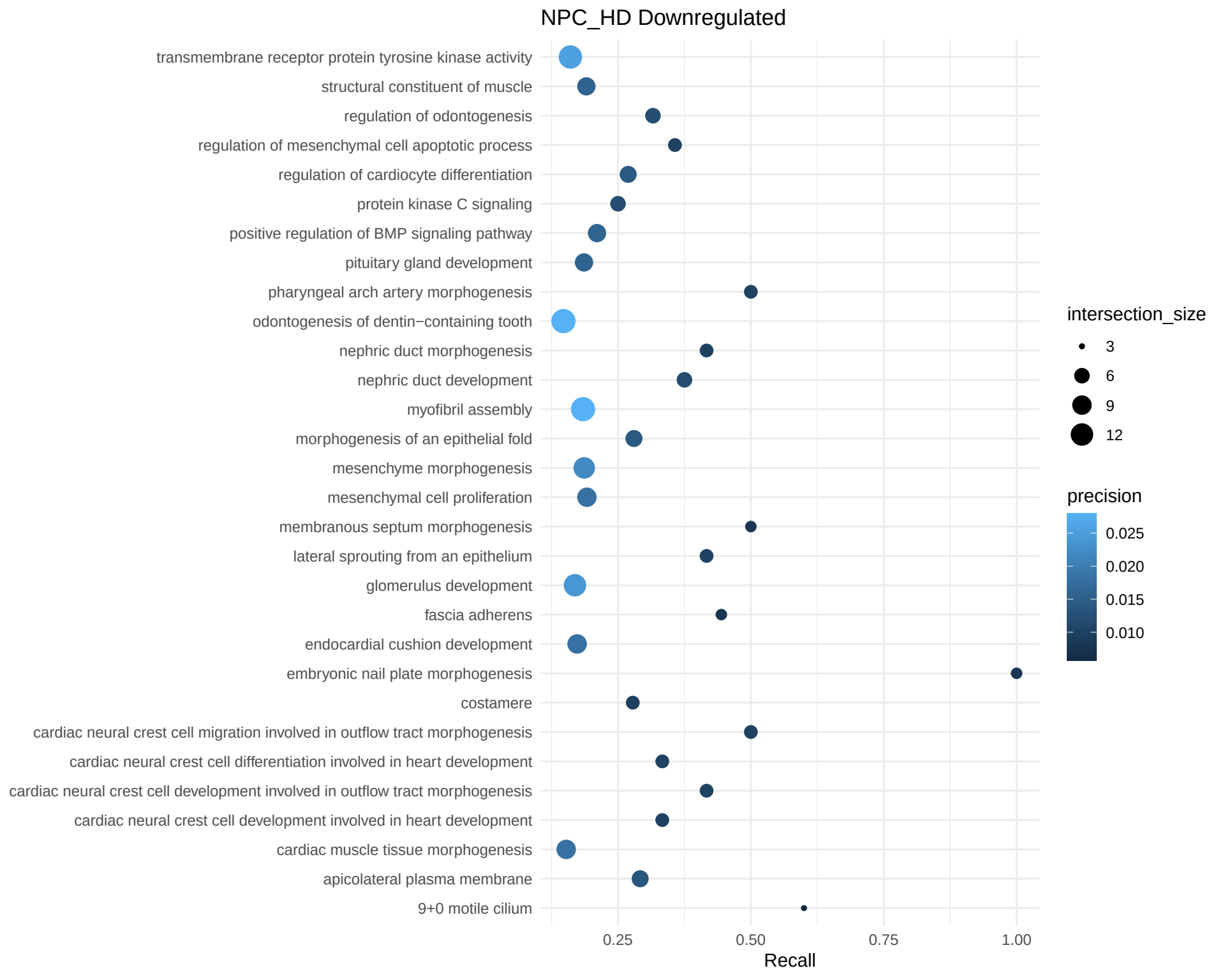

VUSvsWT Upregulated
voltage−gated sodium channel complex
voltage−gated sodium channel activity
voltage−gated potassium channel activity
voltage−gated monoatomic ion channel activity
voltage−gated monoatomic cation channel activity
voltage−gated channel activity
terminal bouton
synaptic vesicle cycle
synaptic transmission, glutamatergic
synaptic transmission involved in micturition
sodium channel complex
signal release from synapse
sensory perception of sound
sensory perception of mechanical stimulus
response to nicotine
potassium ion transmembrane transporter activity
potassium channel activity
postsynaptic intermediate filament cytoskeleton
perikaryon
neurotransmitter transport
neurotransmitter secretion
neuronal cell body
neuron projection terminus
membrane depolarization
G protein−coupled receptor complex
distal axon
cell body
axon terminus
amacrine cell differentiation
action potential
0.25
0.50
0.75
1.00
Recall
intersection_size
10
20
30
40
50
precision
0.075
0.050
0.025

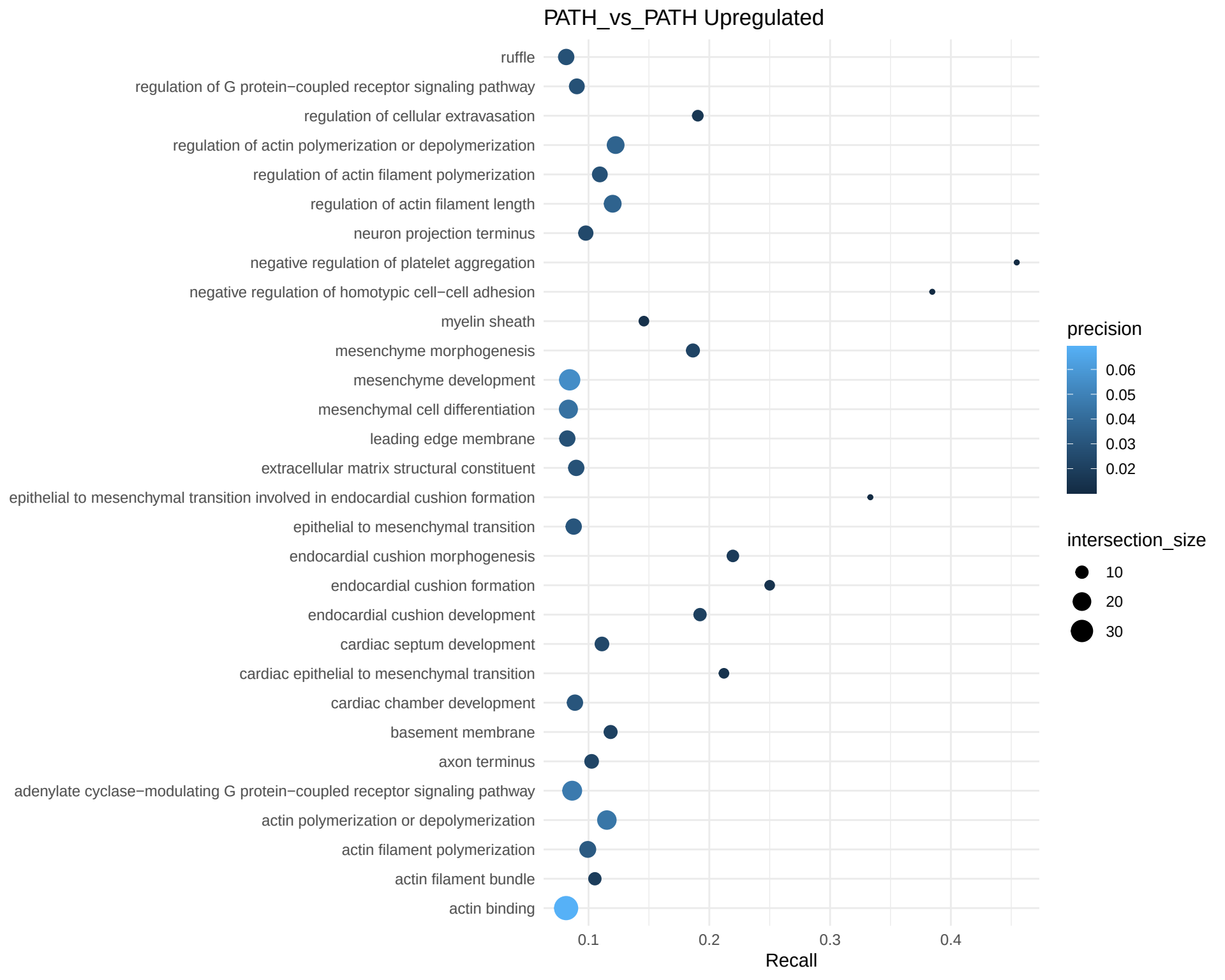

VUSvsWT Downregulated
zonula adherens
structural constituent of muscle
regulation of vascular permeability
regulation of mesenchymal cell apoptotic process
regulation of animal organ morphogenesis
prostatic bud formation
prostate gland morphogenesis
post−anal tail morphogenesis
pituitary gland development
negative regulation of mesenchymal cell apoptotic process
myosin II complex
myosin complex
morphogenesis of an epithelial fold
morphogenesis of an epithelial bud
microfilament motor activity
metanephric mesenchyme development
mesenchymal cell apoptotic process
lymph vessel development
lens morphogenesis in camera−type eye
lateral sprouting from an epithelium
frizzled binding
epithelial cell−cell adhesion
costamere
cell proliferation involved in kidney development
cell differentiation involved in kidney development
branching involved in ureteric bud morphogenesis
apicolateral plasma membrane
animal organ formation
actinin binding
9+0 motile cilium
0.2
0.3
0.4
0.5
0.6
Recall
precision
0.025
0.020
0.015
0.010
intersection_size
3
6
9
12
15

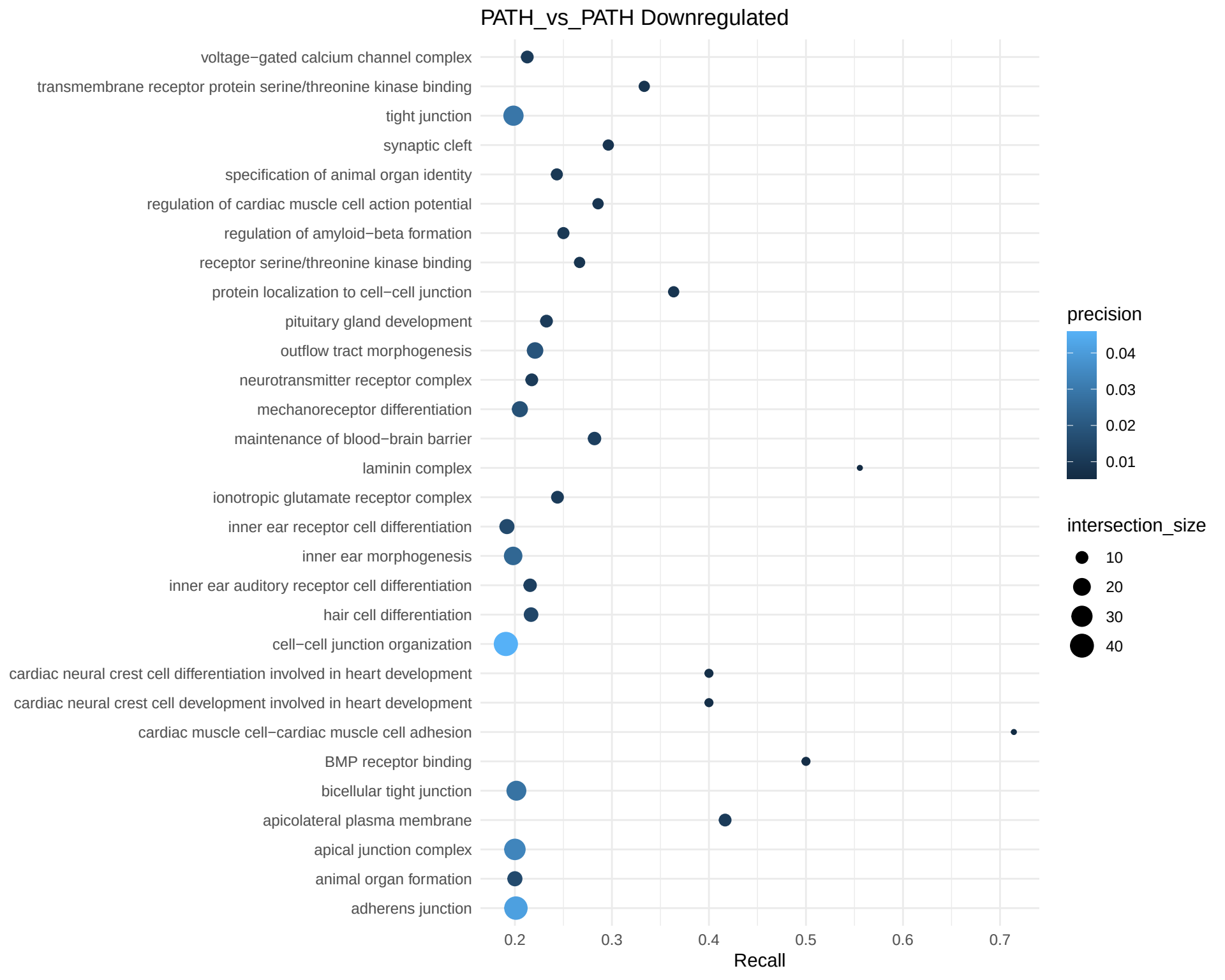

PATH2vsWT Upregulated
synaptic vesicle localization
synaptic transmission, glutamatergic
response to nicotine
regulation of monocyte chemotactic protein−1 production
regulation of dopamine secretion
regulation of catecholamine secretion
positive regulation of calcium ion−dependent exocytosis
parallel fiber to Purkinje cell synapse
neuron projection membrane
negative regulation of lymphocyte migration
negative regulation of leukocyte migration
myelin sheath
monocyte chemotactic protein−1 production
mesenchyme morphogenesis
mesenchymal cell proliferation
main axon
extrinsic component of cytoplasmic side of plasma membrane
excitatory synapse
embryonic cranial skeleton morphogenesis
dopamine transport
dopamine secretion
cochlea development
cell body membrane
catecholamine secretion
behavioral response to nicotine
axon terminus
axolemma
autonomic nervous system development
activin binding
actin filament depolymerization
0.3
0.4
0.5
0.6
0.7
Recall
intersection_size
10
15
20
25
precision
0.020
0.015
0.010
0.005


PATH2vsWT Downregulated
voltage−gated potassium channel activity involved in cardiac muscle cell action potential repolarization
structural molecule activity conferring elasticity
slit diaphragm
regulation of prostatic bud formation
regulation of mesenchymal cell apoptotic process
radial spoke
proximal dendrite
protein binding involved in heterotypic cell−cell adhesion
prostatic bud formation
outer dynein arm
neuroendocrine cell differentiation
nephric duct morphogenesis
nephric duct development
morphogenesis of an epithelial fold
morphogenesis of an epithelial bud
mesenchymal cell apoptotic process
lateral sprouting from an epithelium
laminin complex
glomerular epithelial cell development
filtration diaphragm
extracellular matrix constituent conferring elasticity
epithelial cell−cell adhesion
chemorepellent activity
cell adhesive protein binding involved in bundle of His cell−Purkinje myocyte communication
cardiac muscle cell−cardiac muscle cell adhesion
bundle of His cell−Purkinje myocyte adhesion involved in cell communication
axonemal dynein complex
axonemal central apparatus assembly
axonemal B tubule inner sheath
alpha−catenin binding
0.6
0.8
1.0
Recall
intersection_size
5
7
9
11
13
precision
0.007
0.006
0.005
0.004
0.003


NPC_HD Upregulated
synaptic vesicle recycling
synaptic vesicle membrane
synapse maturation
response to nicotine
response to cocaine
regulation of dopamine secretion
regulation of catecholamine secretion
regulation of amine transport
presynaptic membrane
postsynaptic intermediate filament cytoskeleton
neurofilament
learning
growth cone
GABA−ergic synapse
extrinsic component of plasma membrane
extrinsic component of cytoplasmic side of plasma membrane
exocytic vesicle membrane
excitatory synapse
dopamine transport
dopamine secretion
distal axon
dense core granule localization
catecholamine transport
catecholamine secretion
behavioral response to nicotine
amine transport
adult behavior
acetylcholine−gated monoatomic cation−selective channel activity
acetylcholine−gated channel complex
acetylcholine receptor activity
0.2
0.4
0.6
Recall
precision
0.06
0.04
0.02
intersection_size
3
6
9
12
15


NPC_HD Downregulated
transmembrane receptor protein tyrosine kinase activity
structural constituent of muscle
regulation of odontogenesis
regulation of mesenchymal cell apoptotic process
regulation of cardiocyte differentiation
protein kinase C signaling
positive regulation of BMP signaling pathway
pituitary gland development
pharyngeal arch artery morphogenesis
odontogenesis of dentin−containing tooth
nephric duct morphogenesis
nephric duct development
myofibril assembly
morphogenesis of an epithelial fold
mesenchyme morphogenesis
mesenchymal cell proliferation
membranous septum morphogenesis
lateral sprouting from an epithelium
glomerulus development
fascia adherens
endocardial cushion development
embryonic nail plate morphogenesis
costamere
cardiac neural crest cell migration involved in outflow tract morphogenesis
cardiac neural crest cell differentiation involved in heart development
cardiac neural crest cell development involved in outflow tract morphogenesis
cardiac neural crest cell development involved in heart development
cardiac muscle tissue morphogenesis
apicolateral plasma membrane
9+0 motile cilium
0.25
0.50
0.75
1.00
Recall
intersection_size
3
6
9
12
precision
0.025
0.020
0.015
0.010


PATH_vs_PATH Upregulated
ruffle
regulation of G protein−coupled receptor signaling pathway
regulation of cellular extravasation
regulation of actin polymerization or depolymerization
regulation of actin filament polymerization
regulation of actin filament length
neuron projection terminus
negative regulation of platelet aggregation
negative regulation of homotypic cell−cell adhesion
myelin sheath
mesenchyme morphogenesis
mesenchyme development
mesenchymal cell differentiation
leading edge membrane
extracellular matrix structural constituent
epithelial to mesenchymal transition involved in endocardial cushion formation
epithelial to mesenchymal transition
endocardial cushion morphogenesis
endocardial cushion formation
endocardial cushion development
cardiac septum development
cardiac epithelial to mesenchymal transition
cardiac chamber development
basement membrane
axon terminus
adenylate cyclase−modulating G protein−coupled receptor signaling pathway
actin polymerization or depolymerization
actin filament polymerization
actin filament bundle
actin binding
0.1
0.2
0.3
0.4
Recall
precision
0.06
0.05
0.04
0.03
0.02
intersection_size
10
20
30


PATH_vs_PATH Downregulated
voltage−gated calcium channel complex
transmembrane receptor protein serine/threonine kinase binding
tight junction
synaptic cleft
specification of animal organ identity
regulation of cardiac muscle cell action potential
regulation of amyloid−beta formation
receptor serine/threonine kinase binding
protein localization to cell−cell junction
pituitary gland development
outflow tract morphogenesis
neurotransmitter receptor complex
mechanoreceptor differentiation
maintenance of blood−brain barrier
laminin complex
ionotropic glutamate receptor complex
inner ear receptor cell differentiation
inner ear morphogenesis
inner ear auditory receptor cell differentiation
hair cell differentiation
cell−cell junction organization
cardiac neural crest cell differentiation involved in heart development
cardiac neural crest cell development involved in heart development
cardiac muscle cell−cardiac muscle cell adhesion
BMP receptor binding
bicellular tight junction
apicolateral plasma membrane
apical junction complex
animal organ formation
adherens junction
0.2
0.3
0.4
0.5
0.6
0.7
Recall
precision
0.04
0.03
0.02
0.01
intersection_size
10
20
30
40